

Homework 2: Ensemble Learning for Imbalanced Classification

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1. Introduction

This report studies ensemble-learning techniques for a highly imbalanced COVID-19 severity classification problem. The objective is not simply to maximize overall accuracy, but to determine which learning strategy can identify the rare severe cases while maintaining acceptable performance on the majority class.

Three approaches are investigated. First, a Hellinger Distance Decision Tree (HDDT) is implemented from scratch. Second, bagging is combined with majority-class undersampling and evaluated with both HDDT and a standard decision tree as base learners. Third, AdaBoost.M1 is implemented with decision stumps and combined with SMOTE to study whether synthetic minority oversampling can improve severe-case detection.

All methods are evaluated across 10 independent stratified train-test splits using accuracy, precision, recall, F1-score, AUC-ROC, and G-mean. Particular attention is given to recall, F1-score, and G-mean because the positive class is rare and ordinary accuracy can hide poor minority-class performance.

2. Dataset and Preprocessing

The dataset contains 1,585 observations with 40 clinical and demographic input features and a binary target variable named Label. Severe cases are encoded as +1 and non-severe cases as -1. Only 100 observations belong to the severe class, compared with 1,485 non-severe cases. Therefore, the approximate imbalance ratio is 15:1.

This imbalance creates an important modeling difficulty. A classifier that predicts nearly every patient as non-severe can obtain a high accuracy while being practically ineffective for detecting severe cases. For this reason, the evaluation focuses on minority-sensitive measures in addition to accuracy.

2.1 Missing Values

Missingness is substantial in several clinical variables. Features such as headache, dizziness, paresis, plexus, Chest pain, and inflammation/skin lesions have missing rates above 80%, while Stomachache, nausea, vomiting, diarrhea, and loss of appetite have missing rates around 64.5%.

Median imputation is used for age because it is robust to outliers. Categorical and binary variables are imputed with their most frequent value. Importantly, the imputation parameters are estimated using the training set only. This prevents information from the held-out test set from leaking into model training.

2.2 Correlation-Based Feature Selection

Kendall rank correlation is computed on the training data to identify redundant variables. When two features have an absolute Kendall correlation greater than 0.5, the feature with the weaker absolute relationship to the target is removed. One documented example is vomiting, which is strongly correlated with nausea ($|\tau| = 0.766$). This procedure reduces redundancy while attempting to preserve variables that are more informative for severity prediction.

2.3 Train/Test Protocol and Scaling

A stratified 70/30 train-test split is used so that the rare severe class is represented in both subsets. Continuous variables are standardized using StandardScaler, with the scaler fitted only on the training data and then applied to the test data. For repeated evaluation, random seeds 0 through 9 are used and final performance is summarized

using mean and standard deviation.

3. Task 1: Hellinger Distance Decision Tree (HDDT)

The HDDT is implemented from scratch. Unlike Gini impurity or entropy, the Hellinger splitting criterion normalizes class counts by the total number of positive and negative samples. This makes the criterion less sensitive to prior class probabilities and is therefore attractive for imbalanced classification.

At every node, candidate thresholds are evaluated and the split with the largest Hellinger score is selected. Tree growth stops when the node is pure, the node size falls below `min_samples_split = 10`, the maximum depth is reached, or no valid split is available. A leaf predicts the majority class and uses the fraction of positive observations as its positive-class probability.

3.1 Default HDDT Performance

Metric	Mean	Std. Dev.
Accuracy	0.9109	0.0138
Precision (+1)	0.1562	0.0945
Recall (+1)	0.0833	0.0503
F1-score (+1)	0.1023	0.0564
AUC-ROC	0.5846	0.0186
G-mean	0.2710	0.0868

The default HDDT reaches 91.09% accuracy, but this apparently strong result is dominated by the majority class. Minority recall is only 8.33%, meaning that most severe cases are missed. The low G-mean of 0.2710 confirms that sensitivity and specificity are poorly balanced. Therefore, the result demonstrates why accuracy should not be interpreted alone for this dataset.

3.2 Pruning Experiment

Tree depth was varied over shallow and deeper configurations. At depths 2-4, the classifier essentially never predicts the severe class, producing zero recall and zero G-mean. As depth increases, recall and G-mean improve. The full-depth tree reaches the strongest reported G-mean among the tested HDDT depths. This suggests that the rare positive cases require a relatively detailed partition of the feature space before they can be isolated.

Reducing `min_samples_split` from 10 to 5 or 2 also improves G-mean modestly, to approximately 0.30. Allowing smaller nodes gives the model more opportunities to isolate rare severe samples, although the improvement is not large enough to remove the overall majority bias.

An additional experiment examined the effect of dropping features with more than 80% missingness. Performance became slightly worse, with G-mean decreasing from approximately 0.27 to 0.25. This indicates that even heavily imputed features may still contain useful predictive information.

4. Task 2: Bagging with Undersampling

Bagging with undersampling explicitly addresses class imbalance before fitting each ensemble member. In every iteration, all minority samples are bootstrapped with replacement, while an equal number of majority samples are selected without replacement. The resulting balanced subset contains approximately 200 observations, and a base classifier is trained on it. Final class predictions are obtained by majority voting across the ensemble.

4.1 Effect of Ensemble Size

T	Accuracy	Precision	Recall	F1	AUC-ROC	G-mean
11	0.3250	0.0751	0.8400	0.1371	0.6371	0.4615
31	0.2819	0.0735	0.8867	0.1355	0.6454	0.4439

T	Accuracy	Precision	Recall	F1	AUC-ROC	G-mean
51	0.2708	0.0734	0.9033	0.1357	0.6492	0.4426
101	0.2555	0.0730	0.9167	0.1350	0.6560	0.4191

With HDDT as the base learner, increasing T raises severe-case recall from 84.0% at T=11 to 91.67% at T=101. However, precision stays around 7%, accuracy falls, and G-mean decreases from 0.4615 to 0.4191. Thus, larger ensembles identify more positives but also create many false positives. According to G-mean, T=11 is the strongest of these HDDT bagging configurations.

4.2 Base Learner Comparison

Base learner	Accuracy	Precision	Recall	F1	AUC-ROC	G-mean
Standard DT	0.5345	0.0845	0.6233	0.1474	0.6398	0.5501
HDDT	0.2708	0.0734	0.9033	0.1357	0.6492	0.4426

The choice of base learner has a major effect. HDDT produces very high recall (90.33%) but low accuracy and G-mean, indicating aggressive positive predictions. The standard decision tree produces lower recall (62.33%) but a much better G-mean of 0.5501 and higher F1-score. Since undersampling already balances the data seen by each base classifier, Hellinger loses part of its theoretical advantage, while ordinary Gini-based splitting becomes more suitable.

5. Task 3: AdaBoost.M1 and SMOTE

AdaBoost.M1 trains weak classifiers sequentially. Each round assigns greater importance to samples that were previously misclassified. Decision stumps with `max_depth=1` are used as weak learners. The weighted error of each stump determines its contribution to the final ensemble, and sample weights are updated before the next round.

SMOTE is implemented with `k=5` nearest neighbors. For a minority observation, one minority neighbor is selected and a synthetic point is generated by linear interpolation between the two. Synthetic observations are created until the minority class reaches the size of the majority class. SMOTE is applied only after the train-test split and only to the training set, which is necessary to avoid data leakage.

5.1 AdaBoost With and Without SMOTE

Method	Accuracy	Precision	Recall	F1	AUC-ROC	G-mean
AdaBoost (no SMOTE)	0.9368	0.0000	0.0000	0.0000	0.6113	0.0000
AdaBoost + SMOTE	0.7763	0.0935	0.3100	0.1420	0.5931	0.4774

Without SMOTE, AdaBoost achieves 93.68% accuracy while recall, F1, and G-mean for the severe class are all zero. In practice, this means the classifier obtains high accuracy by predicting the majority class and fails at the actual minority-detection objective. After SMOTE, accuracy decreases to 77.63%, but minority recall rises to 31.0% and G-mean to 0.4774. The lower accuracy is therefore accompanied by a much more useful ability to detect severe cases.

5.2 Number of Boosting Rounds

T	Train error	Test error	F1 (+1)
10	0.2895	0.3639	0.1351
25	0.2077	0.2739	0.1426
50	0.1614	0.2237	0.1420
100	0.1167	0.1651	0.1523

Training error decreases from 28.95% at `T=10` to 11.67% at `T=100`. Test error also decreases, from 36.39% to 16.51%, and the F1-score reaches 0.1523 at `T=100`. No overfitting is observed within the tested range because test error continues to improve as more weak learners are added. The improvement becomes smaller at higher `T`, but there is no clear plateau before 100 rounds.

6. Comparison of the Best Configurations

Method	Best configuration	F1 (+1)	G-mean	AUC-ROC
AdaBoost + SMOTE	<code>T=100</code>	0.1523	0.4463	0.6120
Bagging + Standard DT	<code>T=51, depth=3</code>	0.1474	0.5501	0.6398
Bagging + HDDT	<code>T=11, depth=3</code>	0.1371	0.4615	0.6371
HDDT	<code>depth=None</code>	0.1023	0.2710	0.5846
AdaBoost without SMOTE	<code>T=10</code>	0.0000	0.0000	0.5483

Bagging with undersampling and a standard decision tree gives the highest G-mean (0.5501) and AUC-ROC (0.6398), making it the best balanced method in the experiments. AdaBoost + SMOTE produces the highest F1-score in the summarized best configurations, but its G-mean is lower. Bagging + HDDT obtains very high recall but suffers from excessive false positives. A single HDDT performs considerably worse, and AdaBoost without SMOTE completely fails to identify the minority class.

7. Theoretical Discussion

Hellinger distance versus Gini and information gain. Gini and entropy measure impurity using within-node class proportions and can therefore favor majority-dominated splits. Hellinger compares normalized positive and negative distributions and is designed to be less sensitive to class priors. In the single-tree comparison, HDDT provides a modest improvement in recall and G-mean over a standard decision tree.

Stable, unstable, and weak classifiers. An unstable classifier can change considerably after small changes in the training set; deep decision trees are a typical example. Bagging benefits from this instability because averaging diverse models reduces variance. AdaBoost instead relies on weak learners that perform slightly better than random and can be progressively combined into a stronger model.

Why tree depth is limited in bagging. The bagging base learners use `max_depth=3`. Keeping trees relatively weak encourages diversity among ensemble members and reduces the chance that each model simply memorizes its balanced bootstrap sample. If all ensemble members were effectively identical, majority voting would offer little advantage over a single classifier.

SMOTE and leakage. Applying SMOTE before the train-test split would allow synthetic training points to be influenced by observations that later appear in the test set. This would contaminate evaluation and could artificially increase AUC-ROC and other metrics. The correct procedure is to split first and oversample only the training data.

Clinical interpretation. In a severity-prediction setting, false negatives are especially important because they correspond to severe cases that the model fails to detect. At the same time, excessive false positives reduce precision and can make a classifier impractical. This is why the project uses G-mean and F1-score to evaluate the trade-off instead of selecting a model only by accuracy.

8. Limitations and Possible Improvements

The dataset contains only 100 severe cases, and several variables have extensive missingness. These factors limit how confidently the learned patterns can generalize. The experiments also show low minority precision for several imbalance-aware methods, suggesting that improving recall alone is not sufficient.

Possible extensions include cost-sensitive learning, balanced random forests, alternative SMOTE variants, threshold optimization, probability calibration, cross-validation, hyperparameter search, feature-importance analysis, and additional precision-recall evaluation. Missingness-indicator features could also be explored because the fact that a clinical variable is missing may itself carry information.

9. Conclusion

The experiments demonstrate that class imbalance fundamentally changes how classification results should be interpreted. High accuracy is not evidence of useful performance when the minority class is rare: AdaBoost without SMOTE achieves more than 93% accuracy while detecting no severe cases. Explicit balancing strategies substantially improve minority detection.

Among the tested methods, bagging with majority-class undersampling and standard decision trees provides the best overall balance, achieving $G\text{-mean} = 0.5501$ and $AUC\text{-}ROC = 0.6398$. HDDT improves minority sensitivity compared with ordinary single-tree learning but becomes overly positive when combined with already-balanced bagging samples. AdaBoost benefits strongly from SMOTE and continues to improve through 100 boosting rounds.

Overall, the project highlights three practical lessons: preprocessing must be leakage-safe, model selection for imbalanced data must use minority-sensitive metrics, and the interaction between resampling and the base learner is as important as the choice of ensemble algorithm itself.

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